OpsPilot AWS architecture

Evidence-led offboarding / human approval / simulated access changes

SERVERLESS / us-east-1

aws AWS Cloud

RUNTIME

1 **Requester / approver**
Browser dashboard

HTTPS

2 **Amazon API Gateway**
HTTP API

HTTP routes

3 **AWS Lambda**
API function
Sessions, commands, dashboard assets

Workflow ID

5 **Amazon SQS**
Workflow queue
Messages contain only workflow IDs

SQS trigger

6 **AWS Lambda**
Worker function
Policy, tools and approved execution

Converse API

7 **Amazon Bedrock**
Nova Micro
Tool proposals + receipt summaries

Sessions + workflow state

4 **Amazon DynamoDB**
Workflows, sessions and audit

Checkpoints + receipts

API logs

Worker logs

Scoped execution roles

AWS IAM
Separate API and worker roles
Permissions enforced independently

After 4 receives

8 **Amazon SQS**
Dead-letter queue
Exhausted delivery retries

Queue-depth alarm

9 **Amazon CloudWatch**
Logs + dead-letter alarm
7-day log retention

Human approval binds to the exact action. Execution creates a simulation receipt.

Provisions resources above

DEPLOYMENT

Upload code packages

Developer
AWS SAM CLI

sam deploy

AWS CloudFormation
SAM application stack

Package references

Amazon S3
Deployment artifacts only

AWS Budgets
Account-wide cost alert

50% of $2 / month

Email notification
Budget threshold alert

Runtime flow

Permissions / observability

Deployment only

IAM and Budgets are account-level controls.

Reproducible demo architecture from template.yaml. AWS resources retired November 10, 2026.